Anna-Drishti — Biotechnology Protocol

Jar preparation, incubation, readout, and the science deliverables.

Krittika Parida & Netra R · Smart India Hackathon 2026 · VIT · issued 22 August

THE ONE DEADLINE THAT CANNOT SLIP

The jars must be sealed **today, Saturday**. Grain respiration is a multi-day signal. Sealed by Saturday evening gives roughly 44 hours of incubation, which is just enough to produce a real difference between the two jars. Sealed on Sunday gives nothing measurable by Monday morning. Everything else in this document can slip; this cannot.

WHAT WE ARE TRYING TO PROVE, AND WHY IT IS WORTH THE TROUBLE

Our entire submission currently rests on a simulation. The single most damaging question a judge can ask is *"so none of this is real?"*

Two jars answer that. Same grain, same temperature, differing only in moisture. The wet one respires, produces carbon dioxide and raises the humidity in the headspace; the dry one does not. Feed both readings into our decision engine and it returns **"safe"** for one and **"mould onset in N days, aflatoxin window open"** for the other.

That converts our weakest point into a measurement we took ourselves. It is a small experiment and it is worth more to us than any amount of additional software.

DIVISION OF WORK

Krittika — everything in Parts 1 to 5 (the jars), plus Part 6 (constants sign-off and science slides).
Netra — Part 7, which is entirely separate from the jars and is the most likely thing a judge attacks. Do not let the jar work crowd it out.

Part 1 — Materials and grain selection

Krittika · about 30 minutes including the shop

1.1 · WHICH GRAIN — MAIZE, AND IT IS NOT A FREE CHOICE

Use **whole dried maize kernels**. Not wheat, not rice. Two reasons, the first binding:

The model is already corn. The simulation converts measured humidity into water activity using the modified Chung–Pfost equation with ASAE D245 *shelled corn* coefficients. That equation produces every number we claim. Run the jar on wheat and we are interpreting wheat through corn's sorption curve — and *"your isotherm is for corn but your jar is wheat"* is an easy question to lose to.

Maize is the canonical aflatoxin substrate. *Aspergillus flavus* on maize is the classic pairing in the literature, and maize aflatoxin is a documented Indian problem. Our mycotoxin argument stands on firmer ground with the commodity it was built from.

IF WHAT YOU HAVE IS POPCORN, READ THIS FIRST

Popcorn (*Zea mays* var. *everta*) is still maize, so the experiment works — but two things change, and both must be said out loud rather than quietly ignored:

- **The pericarp is much harder**, so added water penetrates the kernel far more slowly than in dent corn. Across our compressed 44 hours some water may still be sitting on the kernel surface rather than inside it.
- **Surface water reads falsely high.** Free water on the outside drives headspace humidity toward 100 %, which is *not* the grain's true equilibrium water activity. If Jar B reads much above 99 % RH, suspect surface water rather than a real result.
- Our isotherm coefficients are for **shelled dent corn**, not popcorn. Same species, different variety — the sorption curve is close but not identical.

Preferred: 1 kg of feed-grade maize from a poultry feed shop (about ₹30–40/kg) is dent corn, takes up water faster, and matches our isotherm. **But do not let sourcing delay the seal** — popcorn sealed this evening beats dent corn sealed on Sunday. Mitigate by mixing thoroughly and repeatedly, and by using warm (not hot) water at about 40 °C, which penetrates a hard pericarp faster.

IF WE WOULD RATHER THE PROJECT BE WHEAT

Then we change the isotherm coefficients, not the grain. Wheat has its own published Chung–Pfost values; it is a constants edit taking minutes. But it must happen **before Sunday 14:00** and Krittika must sign off the new coefficients. Wheat is more FCI-relevant, maize more aflatoxin-relevant. Our proposal leans mycotoxin, so maize is the recommendation. **Decide before buying.**

1.2 · SHOPPING LIST

ITEM	QTY	NOTES
Whole dried maize kernels ("makka")	**2 kg**	**Provision store, kirana or feed shop. Sound, unbroken, no musty smell. ≈ ₹100**
Glass jars with airtight lids	2	Identical. ~1 L each so 500 g of grain leaves real headspace
Small airtight containers for salt calibration	2	Specimen jars or sample bottles from the lab. Do **not** use the grain jars — those are occupied until Monday. See Part 5B

ITEM	QTY	NOTES
Distilled or clean water	100 mL	Measured with a pipette or syringe
Rubber grommet or silicone sealant	1	**For the sampling port — see 4.2, this matters**

Do not buy: maize flour or atta (no interstitial air, no headspace, meaningless) · sweet corn or fresh cobs (wrong moisture entirely) · anything already visibly mouldy.

Buy 2 kg rather than 1: 500 g × 3 jars, plus grain for the moisture test, plus a margin for a repeat.

From the lab

- Analytical balance, 0.01 g resolution or better
- Drying oven
- Desiccator with active silica gel
- Moisture tins or crucibles, ×3
- Mortar and pestle or a small mill
- Incubator set to 30–35 °C, if one is free
- Nitrile gloves and a dust mask

Part 2 — Determine the starting moisture

Krittika · 2–3 hours, mostly oven time. Start this first, then do Part 3 while it runs.

DO NOT ASSUME 12 %

Bazaar maize runs anywhere from 9 % to 14 %. The water we must add changes by a factor of two across that range, and getting it wrong means the test jar lands at the wrong water activity. Measure first.

2.1 · METHOD — AND ITS HONEST DEVIATION

The reference method for maize is **ASAE S352.2: whole kernel, 103 °C for 72 hours**. We do not have 72 hours. We will use the accelerated ground-sample method and **state the deviation whenever we quote the number.** An acknowledged approximation is defensible; an unacknowledged one is not.

1 Dry the tins

Dry three moisture tins at 130 °C for 1 hour, cool in the desiccator, weigh each empty. Record as `T`.

2 Grind and weigh

Coarsely grind about 60 g of maize. Weigh roughly **15 g** into each tin. Record tin + wet sample as `W1`. Work quickly — ground grain exchanges moisture with room air within minutes.

3 Dry

130 °C for 2 hours, lids off.

4 Cool in the desiccator — do not skip this

Lids on, cool to room temperature in the desiccator, 30–45 minutes. Hot grain weighed in open air absorbs moisture as it cools and reads falsely wet.

5 Reweigh and calculate

Record tin + dry sample as `W2`.

```
moisture % wet basis = (W1 − W2) / (W1 − T) × 100
```

Average the three replicates. If any one differs from the others by more than 0.5 percentage points, discard it and note why.

2.2 · RECORD HERE

REPLICATE	T — EMPTY TIN (G)	W1 — TIN + WET (G)	W2 — TIN + DRY (G)	MOISTURE % WB
1				
2				
3				
Mean				

Method used: .... Deviation from ASAE S352.2 noted: ☐ yes

KEEP A RETAINED SAMPLE

Seal 200 g of the untouched original grain and label it. If we reach nationals we can run the full 72-hour reference method on it and report a properly standardised figure.

Part 3 — Calculate and add water

Krittika · 30 minutes

3.1 · THE CALCULATION

Target for the test jar is **18 % wet basis**. That gives water activity around 0.92 — comfortably above the 0.82 growth floor, respiring hard, and still a realistic wet pocket rather than sodden grain.

```
dry matter = 500 × (1 − start% / 100)
target mass = dry matter / (1 − target% / 100)
water to add = target mass − 500
```

Worked example, starting at 12 %

```
dry matter = 500 × 0.88 = 440.0 g
target mass = 440.0 / 0.82 = 536.6 g
water to add = 536.6 − 500 = 36.6 g ≈ 37 mL
```

3.2 · LOOK-UP TABLE

MEASURED STARTING MOISTURE	ADD PER 500 G FOR 18 %	ADD PER 500 G FOR 20 %
9 %	55 mL	69 mL
10 %	49 mL	63 mL
11 %	43 mL	56 mL
12 %	37 mL	50 mL
13 %	30 mL	44 mL
14 %	24 mL	38 mL

Water to add, Jar B: .... mL   Water to add, Jar C: .... mL

Part 4 — Assemble and seal the jars

Krittika · 45 minutes, then leave them alone

4.1 · THE THREE JARS

JAR	CONTENTS	TARGET A_W	EXPECTED OUTCOME
A control	500 g maize exactly as bought. Nothing added.	≈ 0.65	Below the growth floor. Engine says safe.
B test	**500 g maize + water to 18 % wb**	**≈ 0.92**	**Respiring. Engine predicts mould onset.**
C optional	Only if a third jar appears. 500 g + water to 20 % wb	≈ 0.96	Insurance. Skip if we only have two jars.

With two jars, run A and B. That is the whole experiment — one control, one test. Jar C is insurance against a marginal signal, not a requirement. **Do not sacrifice the control to make a second test jar:** without Jar A there is nothing to compare against and the result is worthless.

4.2 · PREPARE A SAMPLING PORT — DO THIS NOW, NOT MONDAY

OPENING A JAR DESTROYS THE CO_2 MEASUREMENT INSTANTLY

Carbon dioxide accumulated over two days vents in seconds and takes days to rebuild. If we plan to open the jars on Monday to insert a sensor, there is no point running the experiment.

Fix, today: drill a hole in each lid just large enough for the sensor probe or a length of tubing. Fit a rubber grommet, or seal with silicone. Plug it with a bung or a wad of Blu-Tack. On Monday, remove the plug, insert the probe, read, replug.

Humidity is more forgiving. The grain itself is a large moisture reservoir, so headspace humidity re-equilibrates within roughly 15–30 minutes after a brief opening. If the port fails or no CO_2 sensor materialises, water activity is still a valid measurement.

4.3 · PROCEDURE

1. **Label first**

   Jar, target moisture, date and time sealed. Label before filling — three identical jars of maize become unidentifiable fast.

2. **Weigh 500 g into each jar**

   Record actual masses; they will not be exactly 500 g and the calculation should use the real figure.

3. **Add water to B and C**

   Pipette or syringe, added in stages while stirring, not poured in at once. Aim to wet all kernels rather than soaking a few.

4. **Mix thoroughly, then seal**

   Two minutes of turning and stirring. Seal, and plug the port.

5 **Remix at 1 hour and at 3 hours**

Invert and shake each jar. This distributes water into the kernels instead of leaving it on the surface.

6 **Incubate at 30–35 °C**

All three jars at the **same** temperature, undisturbed until Monday.

ONE VARIABLE ONLY

If Jar B sits warmer than Jar A, the comparison is worthless and a judge will say so. Same grain, same mass, same container type, same shelf, same temperature. Moisture is the only thing that differs between them.

BE HONEST ABOUT THE COMPRESSED METHOD

Proper practice is to temper wetted grain cool for 24 hours with periodic mixing so water equilibrates into the kernel *before* incubation begins. We do not have that time. Monday's reading will be near equilibrium but not textbook. **Say so if asked.** It is a far better answer than pretending otherwise, and it costs us nothing — the comparison between jars is still valid because both were treated identically.

Do not inoculate

Ambient spores already present on the kernels are entirely sufficient. Deliberately culturing *A. flavus* requires containment and the mentor's sign-off, and natural colonisation is real data without any of that burden.

4.4 · RECORD

JAR	GRAIN MASS (G)	WATER ADDED (ML)	CALCULATED % WB	TIME SEALED
A		0		
B				
C				

Incubation temperature: .... °C Location: ....

Part 5 — Observation and Monday readout

Krittika · a few minutes Sunday, 30 minutes Monday morning

5.1 · OBSERVATION LOG

Check each jar Sunday morning and Sunday evening. Do **not** open them. Record what is visible through the glass — condensation, discolouration, visible mycelium, clumping.

TIME	JAR A	JAR B	JAR C
Sun morning			
Sun evening			
Mon morning			

Condensation on the inside of Jar B is a good early sign — it means the headspace has reached high humidity, which is exactly what our humidity sensor will measure.

5.2 · MONDAY READOUT PROCEDURE

1 Read the control first

Always Jar A before Jar B. If the sensor drifts or misbehaves, we find out on the jar whose answer we already know.

2 Insert the probe through the port

Remove the plug, insert, reseal around the probe. Minimise the time the port is open.

3 Wait for a stable reading

Humidity typically settles in 5–15 minutes; CO_2 in 1–2 minutes once the sensor has flushed. Record only when the value stops climbing.

4 Record humidity AND temperature together

Water activity is temperature-dependent, and the software needs both. A humidity figure without its temperature is not usable.

5 Type both into the demo

Open `index.html`, find the **Bench reading** panel at the bottom right, enter RH and temperature. It returns water activity, implied moisture, days to mould onset, and whether the aflatoxin window is open.

5.3 · RESULTS

JAR	RH %	TEMP °C	CO_2 PPM	A_W (FROM DEMO)	DAYS TO MOULD (FROM DEMO)
A					
B					
C					

WHAT TO SAY ON STAGE

"This jar is maize we wetted on Saturday morning. The sensor reads 91 % humidity at 31 °C. That is not the simulation — it is our grain, and here is what our engine says about it." Then type it in and let the panel answer. Read the control jar afterwards and let it come back below the growth floor.

5.4 · SAFETY AND DISPOSAL — NOT BOILERPLATE

We are deliberately growing *Aspergillus flavus* on its preferred substrate, warm, for two days. That is an aflatoxin-producing organism and the spores are a respiratory allergen.

- Gloves and a dust mask whenever a jar is open
- Open jars in the lab, never in a demo room, and do not lean over and inhale
- Krittika handles them — this is exactly the mentor-supervised work our proposal commits to
- Keep them **sealed** on Monday. Sealed jars on the table are a good prop; open ones are a hazard and a bad look
- Autoclave before disposal, or seal and dispose per lab protocol. Do not bin an open jar
- Nobody tastes anything

☐ Maize bought ☐ Moisture measured ☐ Ports drilled ☐ Jars sealed ☐ Incubating ☐ Read Monday

Signed: .... Date: ....

Part 5B — Calibrate the sensor against salt standards

Krittika · Sunday, 3–4 hours mostly waiting · needs NaCl and K_2SO_4, both of which we already have

---

A saturated salt solution holds a **known** water activity in the air above it. This is the standard method for checking a humidity sensor, the salts are in the lab, and it costs essentially nothing. It converts *"our sensor says 0.87"* into *"our sensor is calibrated against standards and says 0.87."*

SALT (SATURATED)	A_W AT 25 °C	A_W AT 30 °C	ROLE
Sodium chloride, NaCl	0.753	0.751	Below our 0.82 threshold
Potassium sulphate, K_2SO_4	0.973	0.970	Above it

These two **bracket the 0.82 growth floor**, which is exactly the region our claims depend on. Both are also unusually flat with temperature, so small thermal drift will not spoil the calibration.

PROCEDURE

1 **Make a slurry, not a solution**

Add salt to a little distilled water and keep adding until no more dissolves and there is **visible undissolved solid** at the bottom. That excess solid is what holds the headspace at a fixed water activity. A clear solution with no residue is not saturated, and the calibration is void.

2 **Seal the sensor in with it**

Small airtight container, slurry at the bottom, sensor in the headspace and **not touching the liquid**. Salt water on the sensor will damage it.

3 **Wait for equilibrium**

Two to four hours in a small container; overnight is better. The reading has settled when the firmware's stability flag reads STABLE.

4 **Record and compare**

Repeat for the second salt. If both readings sit within about ±2 % RH of the table, the sensor is good and we say so. If there is a consistent offset, note it and apply it to the jar readings — **a known offset is a calibration; an unknown one is an error.**

SALT	EXPECTED RH %	SENSOR READ RH %	TEMP °C	OFFSET
NaCl	75.3			
K_2SO_4	97.3			

> **WHY THIS IS WORTH A SUNDAY AFTERNOON**
> Everything we claim about water activity rests on a single humidity number. A judge asking *"how do you know your sensor is accurate?"* is asking a fair question, and "it is a good sensor" is not an answer. Two salts and some patience turn that into evidence.

Part 6 — Krittika's other deliverables

These are separate from the jars and equally important

6.1 · SIGN OFF THE CONSTANTS — BY SUNDAY 12:00

You have a separate document, **Anna-Drishti Verification Sheet**, pages 1–3. Fourteen items. Every biological number in the simulation, in plain language, with the model's actual outputs beside it. Tick, correct, and cite a source.

Three of those items are changes made during calibration that specifically need your judgement:

- **Bulk grain moisture 13.5 % → 12.0 %.** Forced, not preference — at 13.5 % the sound grain respires hard enough to mask the spoilage pocket entirely.
- **Aflatoxin temperature ceiling 30 °C → 35 °C.** 25–30 is the reported optimum, not the permissive limit; at 30 the toxin window never opened at all.
- **The growth-rate table** — a_w 0.87 at 30 °C giving mould onset in 17.4 days, and the seven other rows. This is the single most important thing on your pages.

6.2 · SCIENCE SLIDES

Three things, and no more:

1. **Why water activity, not moisture content.** Fungi respond to available water, not total water. This is why our system converts humidity to a_w rather than reporting a moisture percentage.
2. **Why CO_2 leads temperature.** Metabolic gas moves through interstitial air; heat has to conduct through grain, which is an insulator. Our model measures the gap at roughly 0.9 days versus 15 days over the same 40 cm.
3. **Where every constant came from.** One slide, sourced. This is what makes the rest credible.

6.3 · THE QUESTION YOU WILL BE ASKED

"How do you know this reflects real spoilage?"

Your answer has three parts now, and the third is new: the growth model uses published cardinal values for *A. flavus*; the isotherm is a published equation with published coefficients for this commodity; and **we ran it against our own grain and it agreed.** Hold up the jar.

Write your answer out: ....

....

Part 7 — Netra: the inspection baseline

Entirely separate from the jars. This is the most likely thing a judge attacks.

WHY THIS OUTRANKS THE JAR WORK

Our headline claim is *"conventional inspection found it on day 42, we flagged it on day 4."* That claim is only as strong as the model of conventional practice underneath it. If the baseline is pessimistic, the entire comparison collapses under a single question — and a rigged comparison does more damage than no comparison at all. Push back hard on anything below that flatters us.

7.1 · SIGN OFF THE BASELINE MODEL — BY SUNDAY 12:00

Verification Sheet pages 4–5, eight items. The model is coded; these are its assumptions:

ASSUMPTION	CURRENTLY	YOUR CALL
Inspection interval	`every 14 days`	Is fortnightly real for an Indian godown?
Spear reach into a stack	`0.4 m`	**This single number decides the comparison**
Chance of finding a pocket in reach	`60 %`	Too harsh, or too generous?
Surface mould always caught	`yes`	Is certainty fair?
Musty-stack detection route	`added`	Fair, or a fudge? See below

> **THE CHANGE THAT NEEDS YOUR JUDGEMENT MOST**
>
> Our demo pocket sits 1.6 m inside the stack. A spear reaches 0.4 m, so it is **physically unreachable** and the baseline originally read "never detected in 60 days." That is arguably true to FAO guidance — but it reads as rigged.
>
> A second route was added: once a colony is well established the stack smells musty and bags feel warm, so it is noticed at the next routine inspection even without a spear reaching it. That produces day 42. **Decide whether that route is real and whether the timing is fair.**

7.2 · WRITE THE JUSTIFICATION — THIS DOES NOT EXIST YET

The model is coded. The written defence of it is unwritten and only you can produce it. In prose, with citations, it must answer:

- How often inspection happens, and on whose authority
- Which part of a stack a spear can physically reach, and which it cannot
- Why absence of live adults in a sample does not indicate absence of infestation — eggs and larvae develop concealed within kernels
- Why insects concentrate in dust pockets and zones of heating rather than distributing evenly

Ground all four in the FAO guidance already cited in our proposal (references 3 and 4).

7.3 · THE QUESTION YOU WILL BE ASKED

"Isn't your baseline unfairly pessimistic?"

The answer must cite a real inspection interval and a real detection threshold, not an opinion. The strongest version is not a defence of the interval at all — it is that **a spear reaches 0.4 m and the pocket is 1.6 m in.** Inspection is not being modelled as incompetent; it is being modelled as physically unable to reach the location, which is precisely why an interior probe exists.

Write your answer out: ....

....

....

☐ Pages 4–5 signed off ☐ Justification drafted ☐ Sent to Chiranjib ☐ Answer rehearsed out loud

Signed: .... Date: ....